

# **The Role of AI in Enhancing Accuracy of Schizophrenia Diagnosis**

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## Introduction

Schizophrenia is a chronic mental disorder that is characterized by symptoms such as delusions and hallucinations, including perceiving voices that do not exist. Every year, schizophrenia affects about 20 million people worldwide (He et al., 2020). Diagnosis is difficult for schizophrenia because there are no clear biological tests to confidently assess it, meaning clinicians must rely on their clinical judgment. In recent years, however, new developments in machine learning allow AI systems to help doctors examine complex patterns in electroencephalogram (EEG) scans. In this paper, I will examine how AI systems and clinicians can work together to improve schizophrenia diagnosis through EEG analysis, discuss its potential benefits for healthcare systems, and address the ethical and practical challenges that come with using it.

## Discussion and Analysis

### Traditional Methods of Schizophrenia Diagnosis Before AI

Clinicians use a variety of methods to diagnose schizophrenia, including clinical interviews and brain activity scans. During interviews, they observe symptoms such as hallucinations or cognitive dysfunction, often using medical guidelines and scales like the DSM-5 and the PANSS. Brain structure scans like MRI can also be used on patients, mainly to rule out physical causes for the patient's abnormal behaviors (Felman, 2023). Brain activity scans like EEG are also used to detect abnormal patterns in electrical activity that may be associated with the disorder.

## Misdiagnosis of Schizophrenia

Scientists and clinicians have struggled in recent years to accurately diagnose schizophrenia. The misdiagnosis may occur because many mental health conditions share similar behavioral symptoms that may be hard to distinguish during clinical interviews, even with the help of DSM-5 or PANSS. For instance, it is not uncommon for schizophrenia to be confused with bipolar disorder, depression or borderline personality disorder (Rahul et al., 2024). As for physical symptoms, brain structure scans like MRIs can sometimes show physical changes that are present in more than one of these disorders as well. Brain activity scans like the EEG are challenging to interpret due to complex lines and noise, making it difficult to identify clear patterns associated with the disorder (Rahul et al., 2024). Altogether, these challenges show the need for more accurate and practical tools for schizophrenia diagnosis.

## How AI Can Be Utilized to Detect Schizophrenia

AI algorithms can analyze EEG readings and reliably detect brain activity patterns associated with schizophrenia. AI algorithms operate in two ways: machine learning and deep learning models. Traditional machine learning methods, such as random forests, examine specific EEG characteristics that distinguish schizophrenia patients from healthy individuals. Similarly, deep learning models like Convolutional Neural Networks (CNN) analyze raw EEG data and automatically categorize it into different features and hierarchies to detect subtle patterns (Janiesch, 2021). These CNNs have multiple hidden layers of nodes, each possessing different values and weights. For the CNN models to iteratively improve their accuracy, they are trained on validated on multiple datasets, and node values and weights are adjusted after every iteration.

Choosing between these two models can depend on various factors, such as the computational resources available and whether or not the data is preprocessed or not. Unlike machine learning models, deep learning models do not rely on well-selected features by data scientists to analyze data. Deep learning models require more computational resources though, requiring powerful GPUs (Rahul et al., 2024). Both of these algorithms can process large amounts of neural data, identifying patterns that the human eye may miss. However, AI cannot be used in isolation to diagnose schizophrenia, as ethical concerns and algorithmic biases may arise, as explained in more detail later in the essay.

#### What AI-Integrated Schizophrenia Diagnosis Could Look Like

AI can be integrated into the healthcare system to shorten the diagnosis process. Patients often wait weeks or months for evaluations in traditional clinician-only diagnoses; however, they could receive EEG scans at the nearest clinic and immediately receive their AI analysis. These scans could identify potential schizophrenia symptoms and refer patients who are high risk to clinicians. AI can more efficiently analyze large amounts of data at once and find patterns in very noisy data, whereas humans are better at making decisions based on years of real-world experience, and understanding the nuances for each patient's case that AI may have trouble taking into account. By providing early and accurate treatment, AI and clinicians together could prevent severe complications and significantly improve outcomes for schizophrenia patients. This AI-clinician approach demonstrates how AI could improve diagnostics accuracy and speed while maintaining crucial human oversight to ensure the results by AI are reasonable and non-biased. With this saved time, clinicians can focus on other important clinical tasks, such as

conducting research, providing personalized treatment plans for patients, and providing mental health counselling (Alowais et al., 2023).

### Challenges of AI in Schizophrenia Diagnosis

For AI to successfully be integrated into the medical field to assist clinicians, there are many challenges that must be overcome. The first challenge is that AI systems require access to large amounts of sensitive patient data, which contains personal information. If insufficient limitations are placed on these AI systems to protect user data, it could be improperly used, leading to identity theft or breaches of privacy. This sensitive data is more than a patient's name, address, phone number; rather, it is their EEG scans, genetic information, and medical histories. Thus, healthcare providers must strictly follow privacy laws like Canada's PIPEDA to keep patient information safe and ethical (CMPA, 2021).

Another ethical challenge is the issue of accountability. AI cannot be held accountable for making false positive schizophrenia diagnosis for a healthy patient that, if overlooked by a clinician, could have serious consequences. If a human clinician were to make a misdiagnosis, they could be fired or be subject to a lawsuit by the patient. However, these consequences are more complicated to implement on AI. It is difficult to pinpoint the blame on the AI program, or the clinician that overlooked the program's diagnosis, or the developers who coded the AI (Habli et al., 2020). The legal and moral consequences may overcomplicate the implementation of these AI algorithms into health systems until they are accepted and understood by all parties.

An additional challenge is that a large proportion of AI models are trained using data mostly from Western populations, which may not work as well for diverse patient groups. Models may have trouble recognizing subtle differences in brain activity that is specific to Western populations or other populations, thus providing inconsistent diagnosis for patients depending on their ethnic backgrounds. Ethnic minorities in the West tend to be diagnosed with this disorder more often than majorities, even though studies suggest that race does not play a factor (McFarlane & Illes, 2020). To be fair and effective, these systems need training data that represents all populations and be able to recognize the subtle nuances that exist in the brain activities and other characteristics between different populations.

A practical challenge is that AI tools require powerful computers and stable internet, which are often lacking in rural areas in the world. Hospitals can face increased costs and logistic issues when integrating these new technologies. These AI systems need an increased level of computing power to work properly, which could slow its widespread use. These challenges do not mean AI cannot help with schizophrenia diagnosis, but they do show what needs to be fixed before these tools can be used practically in all parts of the world.

There is no doubt that AI is still developing and partially untested. As a result, it will take time for clinicians and patients to have adequate trust for these AI algorithms, as AI will inevitably make mistakes that could undermine its reliability. Many psychiatrists express that they are skeptical of AI systems in the medical field, despite AI showing high accuracy results, because these systems do not have empathy or complex decision-making skills (Harvard Medical School,

2023). This is the reason why a hybrid system must exist so that there is a human aspect of the process.

## Conclusion

In conclusion, the integration of AI into the diagnosis of schizophrenia has the potential to transform healthcare by improving diagnostic accuracy, speeding up the evaluation process, and saving time for clinicians. The ability of AI algorithms to analyze complex data from EEG scans quickly and precisely can enhance the diagnostic process, but it cannot replace the critical role of clinicians. The ideal model is thus the hybrid model, where AI assists but does not replace human expertise, ensures that diagnoses are a combination of advanced technology and real-world experience. Before successful implementation, challenges such as data privacy and accountability in AI training models must be addressed first to ensure that AI does not misuse patient data. If implemented responsibly and practically, this hybrid model can ensure that every patient, no matter where they are in the world, receives accurate, early, and personalized care.

## Bibliography

1. Alowais, S. A., et al. (2023). Revolutionizing healthcare: The role of artificial intelligence in clinical practice. *BMC Medical Education*.  
<https://bmcmmededuc.biomedcentral.com/articles/10.1186/s12909-023-04698-z>

2. Canadian Medical Protective Association. (2021, January). *Protecting your patient's personal health information*. Retrieved from <https://www.cmpa-acpm.ca/en/advice-publications/browse-articles/2011/physician-interactions-with-police>
3. Felman, A. (2023). Does schizophrenia show on a brain scan? Medical News Today. <https://www.medicalnewstoday.com/articles/does-schizophrenia-show-on-a-brain-scan#structural-differences>
4. Habil, I., Lawton, T., & Porter, Z. (2020). Artificial intelligence in health care: Accountability and safety. *Bulletin of the World Health Organization*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7133468/>
5. He, H., et al. (2020). Trends in the incidence and DALYs of schizophrenia. *Epidemiology and Psychiatric Sciences*. <https://pubmed.ncbi.nlm.nih.gov/31928566/>
6. Janiesch, C., et al. (2021). Machine learning and deep learning. *Springer Nature Link*. <https://link.springer.com/article/10.1007/s12525-021-00475-2>
7. McFarlane, J., & Illes, J. (2020). Neuroethics at the interface of machine learning and schizophrenia. *National Library of Medicine*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7371680/>
8. Harvard Medical School (2023). How artificial intelligence is disrupting medicine. <https://postgraduateeducation.hms.harvard.edu/trends-medicine/how-artificial-intelligence-disrupting-medicine-what-means-physicians>
9. Rahul, J., et al. (2024). A systematic review of EEG-based automated schizophrenia classification. *Frontiers in Human Neuroscience*. <https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2024.1347082/full>



